

🟡 CHAPTER 5 — 💪 CHECKING AN ARMSTRONG NUMBER

🟡 SLIDE 1 — 🧠 WHAT IS AN ARMSTRONG NUMBER?

🌟 MEET ARMSTRONG NUMBERS!

Some numbers have a special relationship between their **digits and the original number**.

For example:

153

Take each digit and raise it to the power of 3.

None

$$1^3 = 1$$

$$5^3 = 125$$

$$3^3 = 27$$

Now add them:

None

$$1 + 125 + 27 = 153$$

The answer is the **same as the original number!**



Simple Definition

An Armstrong number is a number equal to the sum of the cubes of its digits.



LOOK AT THIS!

None

153

$$1^3 + 5^3 + 3^3$$

$$1 + 125 + 27$$

$$= 153$$

So:

153 → Armstrong Number ✓



EASY ANALOGY

Think of each digit as having its own **power box** 📦

$$1 \rightarrow 1^3$$

$$5 \rightarrow 5^3$$

$$3 \rightarrow 3^3$$

Then put all the results together:

$$1 + 125 + 27 = 153$$



REMEMBER!

Cube each digit → Add them → Compare with original



SLIDE 2 — 🔍 GETTING THE DIGITS



HOW DO WE GET EACH DIGIT?

We can use the same technique we learned in **Accessing Digits**.

% 10 → Get the last digit 🔍

/ 10 → Remove the last digit ÷

EXAMPLE

Let's take:

153

STEP 1

None

$$153 \% 10 = 3$$

$$153 / 10 = 15$$

👉 Digit = 3

STEP 2

None

$$15 \% 10 = 5$$

$$15 / 10 = 1$$

👉 Digit = 5

STEP 3

None

$$1 \% 10 = 1$$

$$1 / 10 = 0$$

👉 Digit = 1

 **SO WE GET:**

None

153

↓

3 → 5 → 1

Now we can work with each digit.

 **MEMORY TRICK**

% 10 → GET 🔍

/ 10 → REMOVE ÷

🟡 SLIDE 3 — 💪 CUBING EACH DIGIT

🤔 WHAT DO WE DO WITH EACH DIGIT?

For an Armstrong number, we **cube each digit**.

 LOOK AT THIS!

For 153:

None

$$3^3 = 27$$

$$5^3 = 125$$

$$1^3 = 1$$

Now add them:

None

$$27 + 125 + 1 = 153$$

🎯 SO:

None

$$\text{Original} = 153$$

$$\text{Sum} = 153$$

They are equal!

👉 153 is an Armstrong Number ✅

🧠 HOW DO WE CUBE A DIGIT IN C?

We can write:

None

```
digit * digit * digit
```

For example:

None

```
sum = sum + digit * digit * digit;
```



EASY ANALOGY

Think of every digit as entering a **cube machine** ⚙️

None

```
3 → 3 × 3 × 3 → 27
```

```
5 → 5 × 5 × 5 → 125
```

```
1 → 1 × 1 × 1 → 1
```

Then add the results.



REMEMBER!

Each digit → Cube → Add to sum



SLIDE 4 — CHECKING ARMSTRONG IN C



CAN WE DO EVERYTHING USING A LOOP?

Yes! 

We can access every digit and add its cube to `sum`.



BASIC CODE

None

```
int n, original, digit;
int sum = 0;

scanf("%d", &n);

original = n;

while (n != 0)
{
    digit = n % 10;
    sum = sum + digit * digit * digit;
    n = n / 10;
}

if (sum == original)
{
    printf("Armstrong");
}
else
{
    printf("Not Armstrong");
}
```

HOW DOES IT WORK?

For 153:

None

```
153 → digit = 3 → sum = 27
15 → digit = 5 → sum = 152
1 → digit = 1 → sum = 153
```

Finally:

None

```
original = 153
```

```
sum = 153
```

So:

Armstrong 

BIG MEMORY TRICK

% 10 → GET 

digit × digit × digit → CUBE 

sum + cube → ADD 

/ 10 → REMOVE 

sum == original → CHECK 

★ TAKEAWAY

Get → Cube → Add → Remove → Repeat → Compare

SLIDE 5 — ARMSTRONG NUMBER RECAP

WHAT DID WE LEARN?

1. GET THE LAST DIGIT

None

```
digit = n % 10;
```

% 10 → GET 

2. CUBE THE DIGIT

None

```
digit * digit * digit
```

Each digit is cubed.

+ 3. ADD TO THE SUM

None

```
sum = sum + digit * digit * digit;
```

÷ 4. REMOVE THE DIGIT

None

```
n = n / 10;
```

/ 10 → REMOVE ÷

⚖️ 5. COMPARE

None

```
if (sum == original)
```

Same → **Armstrong** ✓

Different → **Not Armstrong** ✗

🧠 ONE BIG MEMORY TRICK!

None

```
% 10
```

↓

```
GET 🔍
```

↓

```
CUBE 💪
```

↓
ADD +
↓
/ 10
↓
REMOVE ÷
↓
REPEAT ↺
↓
COMPARE ⚖

Question

1 Which of the following is an Armstrong Number?

- A) 123
- B) 153
- C) 145
- D) 125

💡 Hint: For 153, calculate $1^3 + 5^3 + 3^3$.

📖 Glossary:

Armstrong Number → A number equal to the sum of the cubes of its digits.

2 Complete the Armstrong Condition

None

```
if (sum ____ original)
{
```

```
printf("Armstrong");  
}
```

💡 Hint: The sum of the cubes must be the same as the original number.

📖 Glossary:

`==` → Checks whether two values are equal.

3 What should `sum` store?

None

```
sum = sum + digit * digit * digit;
```

- A) Only the last digit
- B) The total of all digit cubes
- C) The original number
- D) The number of digits

💡 Hint: Every digit's cube is added to `sum`.

📖 Glossary:

Sum → The accumulated total.

4 What happens first?

For checking an Armstrong number, which happens first?

- A) Compare `sum` and `original`
- B) Cube the digits

- C) Print Armstrong
- D) Compare the answer

 Hint: We need to process the digits before comparing.

 Glossary:

Process → Perform the required operations on the data.

5 Complete the Code

None

```
int n, original, digit;  
int sum = ____;  
scanf("%d", &n);  
original = n;
```

 Hint: The sum must start with an empty total.

 Glossary:

Initialize → Give a variable its starting value.